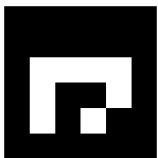
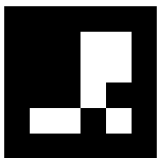


ArUco DICT_4X4_50 — 20 mm

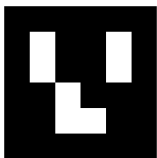
Print at 100%. Turn OFF 'fit to page' / 'shrink oversized pages'.
Measure the black square with a ruler before use: it must read 20 mm across.



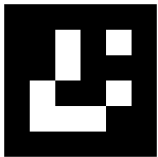
ID 1 · 20 mm



ID 2 · 20 mm



ID 3 · 20 mm



ID 4 · 20 mm



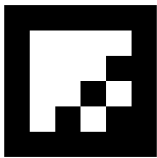
ID 5 · 20 mm



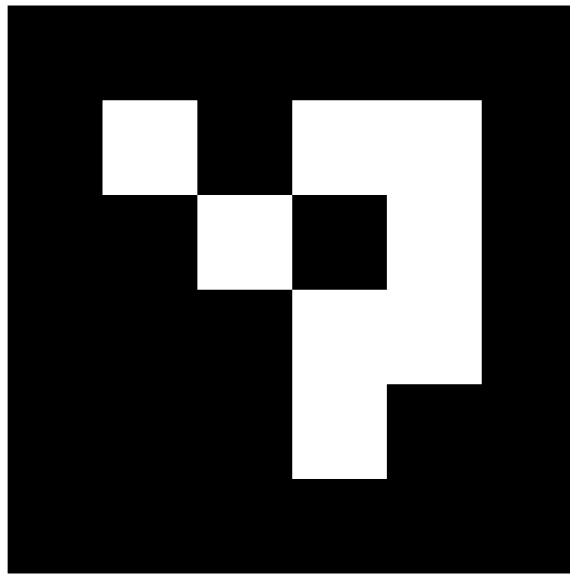
ID 6 · 20 mm



ID 7 · 20 mm



ID 8 · 20 mm



Reference marker · ID 0 · 100 mm

용기 옆 탁자에 평평하게 놓고, 스윙 내내 화면 안에 들어오게 한다.
이 마커가 모든 프레임의 카메라 각도를 준다 — 용기 마커가 놓친 프레임까지.
구겨지거나 휘면 각도가 틀어지므로, 판에 붙이거나 책으로 눌러 둔다.

촬영 조건표 · Shooting grid

Film sweeps, do not take stills. Motion blur and rolling shutter exist only while the camera moves, and they are two of the three things the synthetic study could not model. Posing for each frame removes exactly what the shoot is for.

Lay the large reference marker flat beside the container, in shot throughout. It gives the camera angle for every frame — including the frames where the container's own marker is lost, which are the interesting ones. Recovering the angle from the container's marker would work only where detection worked.

Note the container's diameter: it decides angle tolerance, so a sweep without it cannot be placed on the curve.

Include glossy containers. Specular highlight is the factor the synthetic study could only bound, and matte-only footage would leave it exactly where it was.

Film each scene at two exposures. The model predicts detection turns over on the highlight's peak brightness rather than the light's angular size, so exposure is the first prediction worth testing.

Vary	Levels
Container	지름 기록 · 유광/무광 구분 · 6종
Sweep	카메라를 정면에서 70°까지 천천히 돌리며 20초 녹화 (한 번에 왕복하지 말 것)
Distance	스윙 안에서 30 cm에서 1 m까지 자연스럽게 변화도록
Lighting	실내 확산광 / 직사 조명 (정반사 유도)
Exposure	자동 노출 / 하이라이트 기준 -2 EV

4 sweeps per container (2 x 2), 20 seconds each. Six containers is 8 minutes of recording.

At 30 fps that is about 14,000 frames — thirty times what the stills plan would have produced, and it contains the blur. Drop to one lighting condition if time runs short, but keep both exposures.

Record with every sweep

container ID, container diameter (mm), surface (glossy / matte),
marker ID, marker size (20 mm), lighting, exposure.

Distance and angle are recovered from the footage and need not be written down.
A sweep whose conditions were not written down is not a measurement.

Then: `python scripts/marker_video_eval.py <videos> --ref-id 0 --ref-mm 100`